

Research



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An econometric model for intraday electricity trading

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This paper develops an econometric price model with fundamental impacts for intraday electricity markets of 15-min contracts. A unique dataset of intradaily updated forecasts of renewable power generation is analysed. We use a threshold regression model to examine how 15-min intraday trading depends on the slope of the merit order curve. Our estimation results reveal strong evidence of mean reversion in the price formation mechanism of 15-min contracts. Additionally, prices of neighbouring contracts exhibit strong explanatory power and a positive impact on prices of a given contract. We observe an asymmetric effect of renewable forecast changes on intraday prices depending on the merit-order-curve slope. In general, renewable forecasts have a higher explanatory power at noon than in the morning and evening, but price information is the main driver of 15-min intraday trading.

This article is part of the theme issue ‘The mathematics of energy systems’.

1. Introduction

In recent years, the expansion of renewable energy sources has been forged ahead massively across the globe with a direct impact on electricity markets. As electricity generation from renewable energy sources

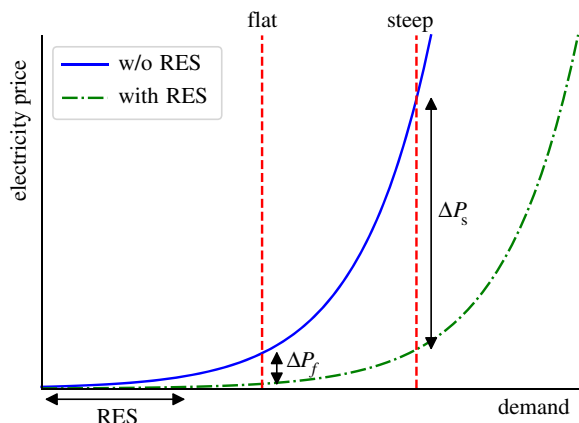


Figure 1. Merit order curve without (blue solid) and with (green dash-dotted) infeed from renewable energy sources (RES) indicating a flat and steep merit-order regime (red dashed) with electricity price changes ΔP_f and ΔP_s , respectively. (Online version in colour.)

cannot be predicted reliably in the long term, that is, days, weeks or months ahead, the future of electricity trading is foreseen in short-term electricity markets. Energy supply companies thus face the challenge of moving towards automatic trading, which requires the identification of trading strategies based on local demand and supply patterns as well as cross-border energy flows. The German market constitutes a pioneer among European electricity markets. Not only is the German market the largest electricity market in Europe in terms of total trading volume [1], but also novel developments and innovations (new contracts, reduction of lead time) are traditionally introduced on the German market first. In Germany, the most short-term electricity market is the continuous intraday market, where hourly and 15-min contracts can be traded until 5 min before the delivery of electricity begins. Intraday electricity markets are designed according to the needs of energy supply companies to balance forecast errors of renewable power generation.

This paper investigates four research questions: (i) Which factors drive the intraday trading of 15-min contracts? (ii) How do forecast changes of renewable power generation affect the intraday trading of 15-min contracts? (iii) Can we identify different regimes on the intraday market where the price formation process behaves differently? (iv) How does intraday trading depend on the time of day?

Like on any financial market, there is a natural desire for pricing models of the basic securities. To build as realistic models as possible, we require *ex-ante* information on the underlying price drivers. More market-specifically, and in light of the design of intraday electricity markets, forecast errors of renewable power production have to be taken into account. [2] deliver the first and only work fulfilling these prerequisites. Our article builds upon their work and develops an econometric price model with fundamental impacts for continuous intraday markets of 15-min contracts.

Our main modelling assumption is that the price formation process on the intraday electricity market depends on the slope of the merit order curve. Figure 1 illustrates a merit order curve with and without the infeed from renewable energy sources (RES). The merit order curve is a nonlinear and convex function of the marginal costs of power plants depending on the generation capacity. Equivalently, since the marginal costs of the last running power plant needed to cover demand set the final electricity price, the merit order curve may be interpreted as a function of the electricity price depending on the electricity demand, too.

Let us consider the merit order curve without renewable power infeed (blue solid). When there is low demand, only relatively cheap power generation technologies such as lignite-fired power plants are needed, whereas if demand is high, more expensive technologies such as gas-fired

power plants might be required to satisfy demand. As exhibited in figure 1, if demand is low, the slope of the merit order curve is relatively small and the market is said to be in a flat merit-order regime; however, if demand is high, the merit-order-curve slope is comparatively large and the market is said to be in a steep merit-order regime.

Let us turn towards the merit order curve with renewable power infeed (green dash-dotted). Since renewable energy sources have zero marginal costs, their infeed shifts the entire merit order curve to the right. As a consequence, if demand is low (flat regime), the electricity price decreases by a small amount ΔP_f ; however, if demand is high (steep regime), the electricity price decreases as well but by a much larger amount $\Delta P_s > \Delta P_f$. We take into account this asymmetric effect of renewable power infeed on electricity prices in our econometric model by incorporating the slope of the merit order curve.

To model the merit order curve, two approaches have been proposed in the academic literature. The first approach models the merit order curve from the supply side via generation capacities and marginal costs of power plants. While generation capacities are given by market transparency data, one has to specify a model for the marginal costs of power plants. [3–5] model the marginal costs as a function of fuel prices, CO₂ emission allowances prices, emission intensities of fuel types, power plant heat rates or thermal efficiencies, and other variable costs. As marginal costs depend on a wide variety of factors, this approach is highly complex and subject to strong modelling assumptions. The second approach models the merit order curve from the demand side via electricity loads, or load forecasts, and electricity prices as proposed by [6,7]. While electricity load is an indicator of total electricity demand, electricity prices are determined by the marginal costs of the price-setting power plant (marginal power plant). As data for both electricity loads and prices exist, this approach does not rely on modelling assumptions. Therefore, we follow the demand-side approach.

The intraday electricity market of 15-min contracts has only recently come into the focus of scientific research. To the best of our knowledge, only three studies existed hitherto. [2] provide the first econometric model for 15-min intraday prices. They model both deviations between day-ahead and last intraday prices as well as continuous intraday prices in a threshold regression context. They provide evidence that 15-min prices respond asymmetrically to intradaily updated renewable forecast errors depending on the proportion of expected demand covered by conventional energy sources. [8] study how the introduction of 15-min contracts has affected the day-ahead and intraday market of hourly contracts. They find that prices of hourly contracts decreased and trading volumes increased. [9] present the first forecasting study of 15-min prices using an elastic net regression model. They conclude that prices in the intraday auction are much easier to forecast than prices in the continuous trading.

This article extends the existing literature along a number of dimensions: first, we explore a novel and unique dataset of high-frequency transaction data and linked fundamental supply and demand data. Intradaily updated forecasts of renewable power generation (solar, wind) constitute the heart of our data collection. These are the same real-time renewable forecasts as available to traders on the intraday market. As such, this is the most extensive dataset used in the empirical literature to study the price formation process on intraday electricity markets. Hence, our dataset allows for a more realistic model specification than proposed in previous research.

Second, we suggest the first econometric price model for 15-min contracts that incorporates the slope of the merit order curve. This is a substantial improvement over [2] as electricity prices react asymmetrically to renewable forecast changes depending on the merit-order-curve slope: if the merit order is steep, electricity prices change more severely in the wake of renewable forecast errors than if the merit order is flat. Moreover, our econometric model solely involves *ex-ante* market knowledge.

Third, and to the best of our knowledge, this is the first work studying the influence of neighbouring 15-min contracts on the price dynamics of a given contract. This is motivated by the fact that adjacent contracts are driven by similar market information.

This paper is organized as follows: in §2, we lay out our dataset and perform an empirical analysis of intraday transaction data of 15-min contracts. In §3, we present our econometric model

as well as the threshold regression. In §4, we calibrate our econometric model to market data and discuss the estimation results. We offer our conclusions in §5.

2. Stylized facts

In this section, we lay out our dataset and empirically analyse transaction prices and trading volumes of 15-min contracts.

(a) Data

We investigate high-frequency trade and linked fundamental data of all 96 15-min contracts traded on the German continuous intraday power market at EPEX SPOT SE. Our observation period spans from 1 January to 31 December 2015. The trade data of 15-min contracts involve transaction prices and trading volumes from the continuous intraday trading session with a 1-min time resolution and are provided by [10]. This, however, does not imply that we observe one transaction every minute, but it may take a multiple of one minute until the subsequent transaction is observed. In case of multiple transactions within the same trading minute, we compute the volume-weighted average price for that minute. The continuous trading session for 15-min contracts opens daily at 16:00 and, in 2015, ends 45 and 30 min before delivery begins, respectively.^{1,2} Furthermore, we use market clearing prices of 15-min contracts traded in the German 15-min intraday auction at EPEX SPOT SE, which takes place daily at 15:00, provided by [13]. The auction data may also be obtained via the R package `emarketcrawler` by [14]. We shall denote a 15-min contract by $HhQq$ with $h = 0, \dots, 23$ and $q = 1, \dots, 4$; e.g. contract H13Q1 refers to the delivery period 13:00–13:15.

As fundamental data, we include intraday wind and solar power forecasts and expected demand. The intraday renewable power forecasts involve intradaily updated forecasts of wind and solar power production in Germany, which are the same real-time renewable forecasts as available to traders on the intraday market. These forecasts are updated every 15 min, where each update contains a forecast time series for the following eight days in a 15-min time resolution, provided by [15]. As such, the intraday renewable power forecasts constitute a unique dataset and, to the best of our knowledge, have solely been analysed by [2]. As an indicator of expected electricity demand, we use the day-ahead total load forecast for each quarter-hour on the following day in Germany, which is published daily at 10:00 and is provided by [16].

(b) Hourly seasonality

(i) Transaction prices

Figure 2 illustrates the volume-weighted average transaction price of 15-min contracts during peak hours and off-peak hours for summer and winter. We identify an hourly seasonality of volume-weighted average prices for both peak and off-peak hours as well as for summer and winter.³ The hourly seasonality exhibits a sawtooth-like shape: for the peak-hour contracts H8Q1–H13Q4, the average price of the first 15-min contract within each hour is the highest and it declines until the last 15-min contract within that hour, which has the lowest average price. Conversely, for contracts H14Q1–H18Q4, the lowest average price is present for the first 15-min contract which increases up to the highest average price for the last 15-min contract in a given hour.

¹EPEX SPOT SE reduced the lead time on the German continuous intraday power market for hourly and 15-min contracts from 45 to 30 min before delivery on 16 July 2015 [11].

²On 14 June 2017, the lead time within the four German control zones was locally further reduced to 5 min before delivery [12].

³The hourly seasonality preserves for unweighted average transaction prices both qualitatively and quantitatively: The seasonal averages of unweighted and volume-weighted average prices differ by roughly 2% only.

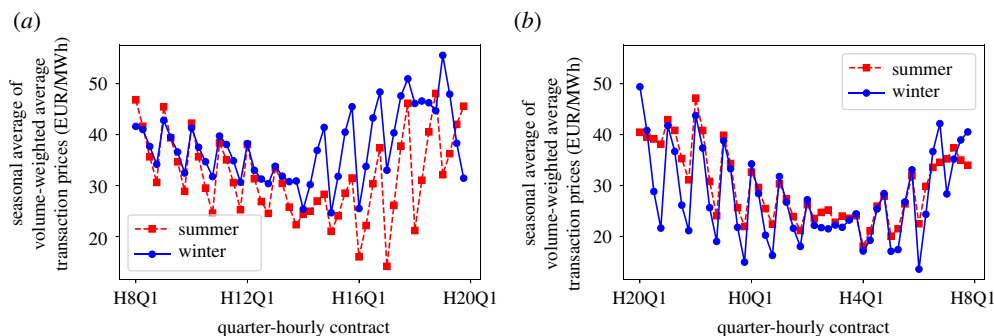


Figure 2. Volume-weighted average transaction price of 15-min contracts during peak hours (a) and off-peak hours (b) averaged over summer (red dashed) and winter (blue solid). (Online version in colour.)

The hourly seasonality pattern and its change around noon may be explained by electricity generation from solar energy: in the first half of the day, the sun rises and less electricity from solar energy is produced during the first quarter-hour when compared with the last quarter-hour within each hour. If a (renewable) electricity supplier sold an hourly contract on the day-ahead market, it has to buy electricity for the first quarter-hour on the intraday market to meet its obligation since less electricity is produced from solar energy than it has sold (buy pressure); thus prices increase. In the last quarter-hour, however, more solar electricity is generated than it has sold on the day-ahead market and so it wants to sell the surplus on the intraday market to avoid entering the balancing energy market (sell pressure); hence prices decrease. In the second half of the day, after the sun has reached its highest level (around 14:00 in Germany), more solar power is generated during the first than in the last quarter-hour in each hour. Thus, there is a sell pressure in the first and a buy pressure in the last quarter-hour of an hour, and the pattern is reversed. The existence of buy and sell pressure is underpinned by the hourly seasonality of trading volumes described in §ii.

Similarly, the sawtooth-shaped hourly seasonality of volume-weighted average prices is found during off-peak hours. For contracts H20Q1–H1Q4, the average price of the first and last quarter-hourly contract within an hour is highest and lowest, respectively, while for contracts H4Q1–H6Q4, this is reversed. The hourly seasonality at night stems from the electricity demand profile in conjunction with established hourly delivery positions from the day-ahead auction (see [17] for a detailed discussion).

Overall, during peak hours, average transaction prices are lower in summer than in winter apart from a few exceptions. In the afternoon and evening hours, that is, for contracts H14Q1–H19Q4, we find larger deviations between summer and winter average prices than in morning and noon hours. During off-peak hours, average prices are fairly similar during both seasons most of the time and only slightly lower in winter than in summer.

(ii) Trading volumes

Figure 3 shows the total trading volume of 15-min contracts during peak hours and off-peak hours averaged over the year. We only present the yearly average of total trading volumes as the distinction between summer and winter does not provide additional information. Similar to transaction prices, we observe an hourly seasonality of total trading volumes for both peak and off-peak hours. The hourly seasonality of trading volumes has a U-shape: larger total trading volumes are found for the first and last 15-min contract within each hour of the day, while the second and third 15-min contract in an hour always exhibit lower trading volumes. More specifically, the last 15-min contract entails the largest trading volume, while the second 15-min contract involves the lowest trading volume in an hour. The U-shaped hourly seasonality supports our hypothesis of buy and sell pressure for the marketing of solar power in the first and last quarter-hour during peak hours, respectively.

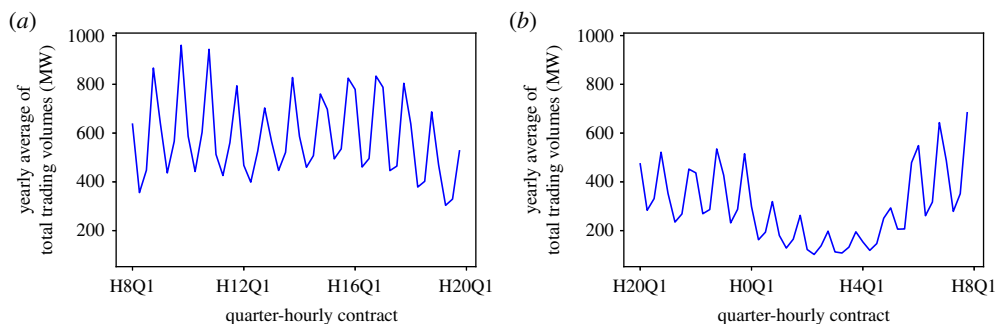


Figure 3. Total trading volume of 15-min contracts during peak hours (a) and off-peak hours (b) averaged over the year. (Online version in colour.)

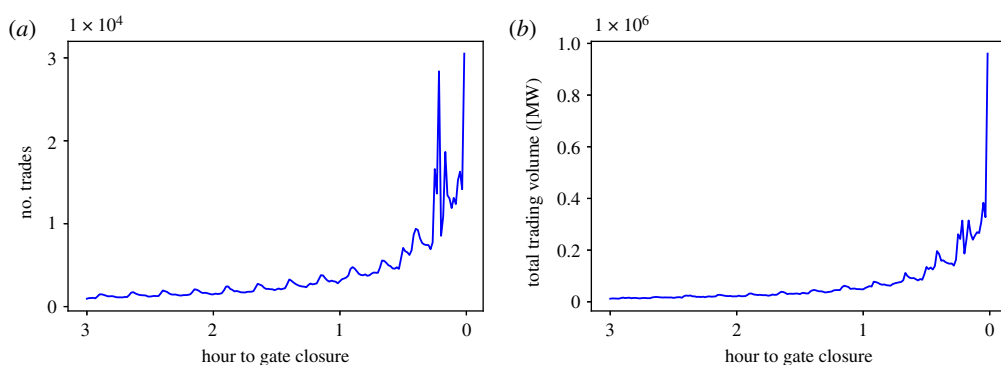


Figure 4. Time evolution of the number of trades (a) and total trading volume (b) through the trading session towards gate closure. (Online version in colour.)

15-min contracts during off-peak hours are generally associated with less trading volume than peak-hour contracts. Total trading volumes are particularly low for contracts H0Q1–H5Q4. However, the U-shaped hourly seasonality of trading volumes is persistent during off-peak hours. The hourly seasonality at night results from the electricity demand profile along with established hourly day-ahead positions [17].

(c) Liquidity evolution

(i) Gate closure

Figure 4 displays the temporal evolution of liquidity of 15-min contracts over the trading session towards gate closure. As measures for market liquidity, we use the number of trades and total trading volume aggregated over all 15-min contracts and all trading sessions. Due to low liquidity far from gate closure, we focus on the last three trading hours prior to gate closure. We note that EPEX SPOT SE reduced the lead time on the German intraday power market for hourly and 15-min contracts from 45 to 30 min before delivery on 16 July 2015 [11]. Thus, to avoid effects due to the shift of lead time, we synchronize our trade time series with respect to gate closure by shifting the trade time stamps of 15-min contracts maturing before 16 July 2015 by 15 min.

The number of trades increases from 952 3 h to gate closure to 3054 1 h to gate closure. Subsequently, the number of trades rises further and more than doubles to 7080 half an hour to gate closure. 15 min to gate closure, the number of trades jumps to 16 593 while 13 min before gate closure it reaches a local maximum of 28 378. The surge of trading activity around 15 min

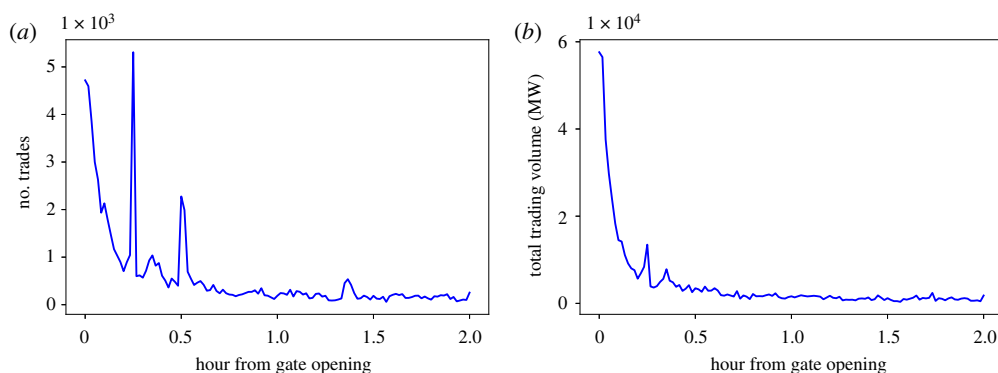


Figure 5. Time evolution of the number of trades (a) and total trading volume (b) after gate opening. (Online version in colour.)

to gate closure may be associated with the fact that a given contract becomes the front 15-min contract. The maximum value of 30 512 trades is observed 1 min to gate closure.

The total trading volume increases from 12 GW to 48 GW between 3 and 1 h to gate closure. Thereafter, trading volume almost triples to 135 GW half an hour to gate closure. 15 and 13 min before gate closure, total trading volume raises to 261 GW and 314 GW, respectively, which coincides with becoming the front 15-min contract. A comparison with the number of trades at these points in time in figure 4 reveals that indeed a vast number of transactions is executed but with comparatively low trading volumes. The total trading volume peaks at 960 GW one minute to gate closure.

Thus, we observe that liquidity of 15-min contracts rises severely within the last trading hour prior to gate closure: on average, roughly 68% of the number of transactions are executed and roughly 74% of the total trading volume is transferred. The reason why the majority of trading takes place close to gate closure is that forecasts of fundamentals, particularly renewable power forecasts, become more and more precise regarding the delivery period of a given 15-min contract. Hence, it is desirable to trade as close to delivery beginning as possible. [18,19] document an increasing liquidity towards gate closure for hourly contracts on the German intraday electricity market, too.

We identify a small but distinct rise in liquidity every 15 min. This is particularly pronounced for the number of trades but present for the total trading volume, too. Consequently, an increased amount of transactions with relatively little trading volume is conducted periodically. We argue that the 15-min periodicity in liquidity originates from newly arriving renewable forecast updates. As described in §a, renewable forecasts are updated in 15-min intervals. New forecasts will not have changed much after 15 min and thus traders only make minor adjustments to their positions by trading little volume. Moreover, we observe that trading activity increases at isolated points in time and dies out immediately until the next increase. Therefore, we conclude that renewable forecast updates are reflected in prices of 15-min contracts within one trading minute.

(ii) Gate opening

Figure 5 illustrates the temporal development of liquidity of 15-min contracts after gate opening at 16:00. The number of trades amounts to 4718 just after gate opening and decreases to 705 12 min after gate opening. 15 min after gate opening, the number of trades reaches its maximum value of 5309, which drains within the next trading minute. Similarly, liquidity jumps to 2276 trades 30 min after gate opening and falls back to its prior level within two trading minutes. Thereafter, liquidity approaches a fairly stable level of 185 trades, on average, between 1 and 2 h after gate opening.

The total trading volume peaks at 58 GW shortly after gate opening and steadily declines to 5.6 GW 12 min after gate opening. 15 min after gate opening, total trading volume rises to 13 GW while it amounts to 3.5 GW 30 min after gate opening. Comparing trading volume and the number of trades at these points in time shows that the surge in trading activity involves relatively little trading volume. Subsequently, the total trading volume keeps a quite constant and low level around 1.5 GW until 2 h after gate opening.

Thus, liquidity of 15-min contracts is high close to gate opening and drops substantially during the first 12 trading minutes. This behaviour may be explained by the fact that market participants initialize their positions. In the sequel, little volume is traded and thereby only minor adjustments are made to the open positions. Generally, liquidity remains poor until the last trading hour prior to gate closure as the forecasts of fundamentals are still relatively inaccurate.

3. Methodology

We aim at modelling the asymmetric response of 15-min intraday electricity prices to explanatory variables, in particular, to renewable forecast changes depending on the slope of the merit order curve. We suggest an extension of the econometric model by [2] to overcome its weaknesses and employ a threshold regression model to calibrate our model to market data.

(a) Extended econometric model

We refine the econometric model by [2] along three dimensions by incorporating supplementarily: (i) the slope of the merit order curve, (ii) price changes of neighboring 15-min contracts, (iii) the 15-min intraday auction price. For a given 15-min contract $i = 1, \dots, 96$, the model specification reads

$$\begin{aligned} \Delta P_t^{(i)} = & \alpha_0^{(i)} + \sum_{\tau=1}^m \alpha_\tau^{(i)} \Delta P_{t-\tau}^{(i)} + \sum_{\substack{j=-n, \\ j \neq 0}}^n \beta_j^{(i)} \Delta P_t^{(i+j)} + \eta_1^{(i)} \xi^{(i)} + \eta_2^{(i)} P^{\text{Auc},(i)} + \eta_3^{(i)} V_t^{(i)} \\ & + \eta_4^{(i)} \Delta w_t^{n,(i)} + \eta_5^{(i)} \Delta w_t^{p,(i)} + \eta_6^{(i)} \Delta s_t^{n,(i)} + \eta_7^{(i)} \Delta s_t^{p,(i)} + \eta_8^{(i)} \sqrt{\Delta t^{(i)}} + \epsilon_t^{(i)}, \end{aligned} \quad (3.1)$$

where $\Delta P_t^{(i)} = P_t^{(i)} - P_{t-1}^{(i)}$ denotes the transaction price change between times t and $t-1$, $\Delta P_t^{(i+j)}$ the last observed price change at time t of neighbouring contract $i+j$, $\xi^{(i)}$ the slope of the merit order curve, $P^{\text{Auc},(i)}$ the 15-min intraday auction price, $V_t^{(i)}$ the trading volume at time t , $\Delta w_t^{n,(i)} = \min(\Delta w_t^{(i)}, 0)$ and $\Delta w_t^{p,(i)} = \max(\Delta w_t^{(i)}, 0)$ negative and positive wind power forecast changes, respectively, where $\Delta w_t^{(i)} = w_t^{(i)} - w_{t-1}^{(i)}$ is the last available wind power forecast change at time t , $\Delta s_t^{n,(i)} = \min(\Delta s_t^{(i)}, 0)$ and $\Delta s_t^{p,(i)} = \max(\Delta s_t^{(i)}, 0)$ negative and positive solar power forecast changes, respectively, where $\Delta s_t^{(i)} = s_t^{(i)} - s_{t-1}^{(i)}$ is the last available solar power forecast change at time t , and $\Delta t^{(i)}$ the interarrival time between two consecutive transactions conducted at times t and $t-1$ of contract i .

The first sum in equation (3.1) covers m lagged price changes $\Delta P_{t-\tau}^{(i)}$, $\tau = 1, \dots, m$, that is, autoregressive terms. To determine the number of lags, we use the partial autocorrelation of price changes and choose $m = 3$. The second sum in equation (3.1) captures the price change $\Delta P_t^{(i+j)}$, $j = -n, \dots, n$, $j \neq 0$, at time t of n 15-min contracts maturing before and n 15-min contracts maturing after contract i . For example, if $n = 2$ and $i = \text{H13Q1}$, the price change of contracts H12Q3, H12Q4, H13Q2, H13Q3 is included. The intraday auction price $P^{\text{Auc},(i)}$ remains constant during the continuous trading session of a given contract i and can be considered as an estimate of the initial price of contract i in the continuous trading. Moreover, we distinguish between positive and negative wind and solar power forecast errors Δw_t^p , Δs_t^p and Δw_t^n , Δs_t^n , respectively, as we expect them to have opposite effects on electricity price changes ΔP_t : positive renewable forecast errors should decrease electricity prices, whereas negative renewable forecast errors should increase electricity prices. We control for the interarrival time Δt since transactions do

not take place at equidistant points in time, but it may take one minute or several hours until the next trade is conducted.

We use the slope of the merit order curve $\xi^{(i)}$ as threshold variable instead of the demand quota compared to [2]. The demand quota is defined as the ratio of the expected demand and the expected conventional capacity. As such, the demand quota quantifies the proportion of expected demand which is expected to be met by conventional power generation capacities; or, put another way, how much the expected conventional capacity does cover expected demand. One weakness of the demand quota is that it does not recognize whether the market is in a flat or steep merit-order-curve regime. Proportionally speaking, a high expected demand and high expected conventional capacity lead to the same value of the demand quota as low demand and low capacity. Another weakness is that the demand quota aggregates expected capacities over all conventional generation technologies. Thus, it loses information on the price-setting power plant and the slope of the merit order curve. We overcome these limitations in our extended econometric model (3.1).

We estimate $\xi^{(i)}$ from empirical intraday auction prices $p^{\text{Auc},(i)}$ and expected demands, or total load forecasts, $\ell^{(i)}$ in the spirit of [6,7]. This approach is reasonable for describing the slope of the merit order curve since the level of intraday auction prices reflects the marginal costs of power plants needed to cover expected demand. [6,7] construct a merit order curve for the electricity spot market as a whole, independent of the contract, from hourly day-ahead prices and hourly load forecasts. We, however, determine a merit order curve for each 15-min contract i individually and thus arrive at a more fine-grained picture. We fit a function $f(\ell)$ to the price-load data and call $f(\ell)$ the empirical merit order curve.⁴ Then, we take the derivative of the empirical merit order curve $f'(\ell) = \frac{df(\ell)}{d\ell}$ and substitute empirical expected demands $\ell^{(i)}$ into $f'(\ell)$ to obtain empirical merit-order-curve slopes $\xi^{(i)} = f'(\ell^{(i)})$.

The slope of the merit order curve $\xi^{(i)}$ remains constant during the trading session of a given contract i . The value of $\xi^{(i)}$ can be determined daily around 15:10, after the publication of the intraday auction results, and hence before continuous trading begins. Thus, $\xi^{(i)}$ indicates whether the market is in a flat or steep merit-order regime before continuous trading begins and the corresponding intraday price model can be chosen *ex-ante*. This is an attractive feature of our econometric model for practical applications.

(b) Threshold regression

We use the threshold regression model introduced by [20] to calibrate our econometric model to market data. The threshold regression model is able to reveal asymmetries in the impact of explanatory variables with respect to a specified threshold variable. The basic concept of the threshold regression involves two steps: first, the entire sample is split into two subsamples, also referred to as groups, classes, or regimes, at a certain threshold value of a designated threshold variable; second, a linear regression model is estimated on each subsample separately.

More formally, suppose we observe the sample $\{y_i, \mathbf{x}_i, q_i\}_{i=1}^n$, where y_i is the dependent variable, $\mathbf{x}_i \in \mathbb{R}^m$ collects the independent variables, and q_i denotes the threshold variable. The threshold regression model reads

$$y_i = \begin{cases} \theta'_1 \mathbf{x}_i + \varepsilon_i, & \text{if } q_i \leq \gamma \\ \theta'_2 \mathbf{x}_i + \varepsilon_i, & \text{if } q_i > \gamma \end{cases},$$

$$= \theta'_1 \mathbf{x}_i \mathbb{1}\{q_i \leq \gamma\} + \theta'_2 \mathbf{x}_i \mathbb{1}\{q_i > \gamma\} + \varepsilon_i, \quad i = 1, \dots, n, \quad (3.2)$$

where γ is the threshold parameter, $\theta_1, \theta_2 \in \mathbb{R}^m$ collect the regression parameters, $\mathbb{1}$ denotes the indicator function, and ε_i is the error term. Thus, the observed sample is split into two subsamples along the threshold variable q_i at a specific value γ . By design, the threshold regression model (3.2) allows the regression parameters in θ_1, θ_2 to vary between the regimes.

⁴[6] call f empirical price load curve.

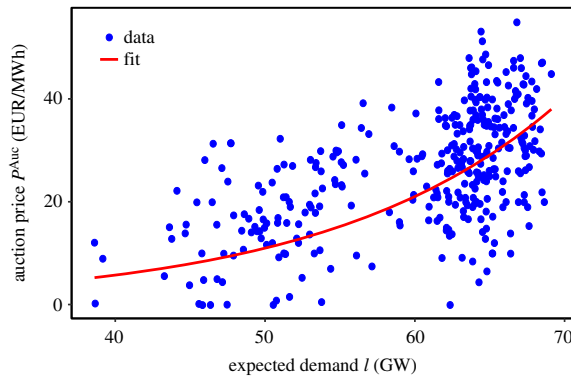


Figure 6. Empirical merit-order-curve function $f(\ell) = e^{a\ell+b}$ (red) fitted to the empirical intraday auction price $p_d^{\text{Auc},(i)}$ as a function of expected demand $\ell_d^{(i)}$ (blue) observed on days $d = 1, \dots, T$, for 15-min contract $i = \text{H13Q4}$. (Online version in colour.)

The regression parameters $\theta_1, \theta_2, \gamma$ can be estimated by least squares and jointly minimize the sum of squared errors function. A detailed description of the estimation of the regression parameters is provided in Section A in the electronic supplementary material. We use the R package `threg` developed by [21] to estimate the threshold regression model.

4. Estimation results

We estimate the parameters of the extended econometric model (3.1) by the threshold regression described in §b for all 96 15-min contracts.⁵ Here a selection of 15-min contracts representative for morning, noon, and evening hours is analysed, that is, H7, H13, H18, Q1–Q4 each. For all contracts, the merit-order-curve slope ξ is used as threshold variable.

(a) Merit-order-curve slope

We estimate the slope of the merit order curve $\xi^{(i)}$ of 15-min contract i based on intraday auction prices $p_d^{\text{Auc},(i)}$ and corresponding expected demands $\ell_d^{(i)}$. As both $p_d^{\text{Auc},(i)}$ and $\ell_d^{(i)}$ are provided in quarter-hourly resolution, we are able to determine a merit order curve for each 15-min contract i individually. Figure 6 depicts a scatter plot of empirical intraday auction prices $p_d^{\text{Auc},(i)}$ versus empirical expected demands $\ell_d^{(i)}$ observed on days $d = 1, \dots, T$, for contract $i = \text{H13Q4}$. We filter out a total of 13 negative auction prices. We observe a positive relationship between intraday auction prices and expected demands: $p_d^{\text{Auc},(i)}$ increases as $\ell_d^{(i)}$ increases. For high levels of demand, more expensive generation technologies are in use which puts upward pressure on prices. In particular, we may identify two clusters: one cluster encompasses expected demands $\ell < 58 \text{ GW}$, and the other cluster comprises $\ell \geq 58 \text{ GW}$. These clusters reflect low and high electricity demand on weekends and weekdays, respectively. Overall, the data points exhibit a fairly wide range of variation which stems from the strongly varying expectation of renewable power infeed.

For the empirical merit order curve $f(\ell)$, we use the exponential function $f(\ell) = e^{a\ell+b}$ following [7]. Of course, other choices for $f(\ell)$ are possible, but we want to keep our model as simple as possible. We fit the exponential function $f(\ell)$ to the empirical intraday auction price $p_d^{\text{Auc},(i)}$ as a function of expected demand $\ell_d^{(i)}$. Technically, we minimize the sum of squared errors between the logarithm of the function $f^\Phi(\ell)$ implied by the choice of a parameter set $\Phi = \{a, b\}$, and the

⁵The estimation time of the threshold regression amounts to roughly one minute per contract.

Table 1. Parameter estimates of a , b of the empirical merit-order-curve function $f(\ell) = e^{a\ell+b}$ for 15-min contracts H7, H13, H18, Q1–Q4 each.

contract	parameter			
	a		b	
	estimate	s.e.	estimate	s.e.
H7Q1	0.029	(0.004)	1.632	(0.268)
H7Q2	0.036	(0.002)	1.412	(0.134)
H7Q3	0.045	(0.002)	0.961	(0.125)
H7Q4	0.045	(0.002)	0.940	(0.130)
H13Q1	0.022	(0.002)	2.194	(0.141)
H13Q2	0.037	(0.003)	1.214	(0.201)
H13Q3	0.045	(0.004)	0.548	(0.217)
H13Q4	0.065	(0.006)	−0.856	(0.389)
H18Q1	0.061	(0.005)	−0.372	(0.313)
H18Q2	0.039	(0.003)	1.246	(0.160)
H18Q3	0.021	(0.002)	2.546	(0.109)
H18Q4	0.013	(0.002)	3.074	(0.140)

logarithm of the empirical intraday auction price $P_d^{\text{Auc},(i)}(\ell_d^{(i)})$,

$$\min_{\Phi} \sum_{d=1}^T | \log(P_d^{\text{Auc},(i)}(\ell_d^{(i)})) - \log(f^{\Phi}(\ell)) |^2.$$

(4.1)

The least-squares fit for contract $i = \text{H13Q4}$ is shown in figure 6. The parameter estimates of a , b for contracts H7, H13, H18, Q1–Q4 each, are reported in table 1.

We observe an hourly seasonality of the parameter estimates $\hat{a}$, $\hat{b}$. For contracts H4Q1–H14Q4, the estimates $\hat{a}$, $\hat{b}$ increase and decrease from the first to the last 15-min contract within an hour, respectively, whereas for contracts H15Q1–H2Q4, the estimates $\hat{a}$, $\hat{b}$ decrease and increase, respectively. Thus, the curvature of the empirical merit order curve $f(\ell)$ grows from Q1 to Q4 in each hour during morning and noon hours, whereas it declines from Q1 to Q4 during afternoon and evening hours. This hourly seasonality may be associated with rising and falling demand in the first and second half of the day, respectively, following human activity [22]. Consequently, in the first half of the day, more expensive power plants are needed to cover demand in the last than in the first quarter-hour in each hour, whereas in the second half of the day, more expensive power plants are operated in the first than in the last quarter-hour per hour.

Eventually, we take the derivative of the fitted merit-order-curve function $f^{\prime \Phi}(\ell) = \text{d}f^{\Phi}(\ell)/\text{d}\ell$ and substitute empirical expected demands $\ell_d^{(i)}$ into $f^{\prime \Phi}(\ell)$ to obtain empirical merit-order-curve slopes $\xi_d^{(i)} = f^{\prime \Phi}(\ell_d^{(i)})$ on days $d = 1, \dots, T$, for 15-min contract i .

(b) Model calibration

The estimation results of all 15-min contracts in hour H13 are presented in table 2, while the estimation results of contracts in hours H7 and H18 may be found in Tables B.1 and B.2 in the electronic supplementary material, respectively. The threshold test shows strong evidence for a threshold effect in the slope of the merit order curve ξ for all contracts in hours H13 and H7,

while in hour H18, it suggests a statistically significant threshold effect for contracts Q2 and Q3.⁶ A merit-order-curve slope of $\xi = 1$ (EUR/MWh)/GW implies that a change of 1 GW in expected demand causes a change of 1 EUR/MWh in the intraday auction price. Regime 1 corresponds to a market in which the merit order curve is flat, whereas Regime 2 reflects a market in which the merit order curve is steep. For all contracts, the total sample is split into subsamples of reasonable size and allows for a sound interpretation of the estimation results. The adjusted R^2 ranges between 11% and 22%.

The estimated coefficients of lagged price changes ΔP_{t-1} , ΔP_{t-2} , ΔP_{t-3} are highly statistically significant and negative in both regimes for all contracts in hours H13 and H7. In hour H18, the higher-order autoregressive terms show less statistical significance. The negative coefficients of lagged price changes suggest mean reversion in the price formation process of 15-min contracts. As such, they reflect the so-called ‘learning effect’ or ‘participant conduct’ [23,24]. Our results confirm previous findings of [2] for morning and evening contracts. For noon contracts, by contrast, they find that autoregressive terms have less explanatory power and intraday trading is primarily driven by renewable forecast changes. We, however, provide evidence of a significant mean reversion effect for noon contracts in 2015.

Overall, in hour H13, the estimated coefficients of price changes of neighboring contracts $\Delta P_t^{(i-2)}, \dots, \Delta P_t^{(i+2)}$ are highly statistically significant and positive in both regimes for all contracts. In hour H7, the estimated coefficients of $\Delta P_t^{(i-2)}, \dots, \Delta P_t^{(i+2)}$ are highly statistically significant and positive in the steep merit-order regime for all contracts as well as in the flat regime for contracts Q3 and Q4. Thus, price changes of neighbouring contracts have strong explanatory power and a positive effect on one another. Moreover, we observe that price changes of the nearest neighbours $i \pm 1$ have a stronger impact on price changes of contract i than price changes of next-nearest neighbours $i \pm 2$.⁷ In particular, for the first 15-min contract Q1 in an hour, price changes $\Delta P_t^{(i+1)}$ of the following contract Q2 have the greatest influence, whereas for the last contract Q4 in that hour, price changes $\Delta P_t^{(i-1)}$ of the preceding contract Q3 exhibit the largest impact. Hence, 15-min contracts within the same hour are the most important price drivers. In hour H18, the coefficients of $\Delta P_t^{(i-2)}, \dots, \Delta P_t^{(i+2)}$ are rarely statistically significant.

The coefficients of the intraday auction price P^{Auc} are not significant for all contracts and regimes. This is not surprising since P^{Auc} can merely be considered as an estimate of the initial price of a 15-min contract in the continuous trading. Thus, it is not expected that P^{Auc} affects continuous trading beyond the first price.

In hour H13, the coefficients of trading volume V_t are significant in the flat merit-order regime for all contracts and in the steep regime for contracts Q1 and Q4. Independent of the regime, they are negative for Q1 and Q2 and positive for Q3 and Q4. This sign profile reflects the joint hourly seasonality of transaction prices and trading volumes described in §b: as more solar power is generated than the hourly average in Q1 and Q2, there is sell pressure at the beginning of the hour and intraday prices decrease. Conversely, buy pressure increases intraday prices at the end of the hour (Q3 and Q4) when below-average solar power is produced. In hour H7, the coefficients of V_t are significant for all contracts and regimes and we observe the opposite sign pattern: for contracts Q1 and Q2, the coefficients are positive, whereas for Q3 and Q4, they turn negative independent of the regime. Thereby, the positive and negative coefficients reflect buy and sell pressure at the beginning and end of the hour, respectively. In hour H18, the coefficients of V_t are not significant. In general, we do not find an asymmetric adjustment of intraday price changes ΔP_t to trading volumes V_t .

Overall, the coefficients of negative wind forecast changes Δw_t^n are significant and negative in the steep merit-order regime for all contracts. The negative sign is economically meaningful as electricity prices should increase if less wind power is forecasted. In hour H13, the coefficients of

⁶We indicate the statistical significance level of the threshold test by superscript asterisks at the threshold variable in the regression tables.

⁷The terms nearest and next-nearest neighbours are motivated from physics where a nearest- and next-nearest-neighbour interaction exists.

Table 2. Estimation results of the extended econometric model (3.1) for intraday price changes ΔP_t of 15-min contracts H13Q1–4.

H13Q1					H13Q2				
variable	regime 1		regime 2		variable	regime 1		regime 2	
	$\xi^{**} \leq 0.689$		$\xi^{**} > 0.689$			$\xi^{***} \leq 0.912$		$\xi^{***} > 0.912$	
	estimate	s.e.	estimate	s.e.		estimate	s.e.	estimate	s.e.
const	−1.504	(1.643)	−0.867	(0.911)	const	−0.201	(1.225)	0.304	(0.734)
ξ	2.610	(2.672)	1.021	(1.054)	ξ	0.931	(1.622)	−0.371	(0.548)
ΔP_{t-1}	−0.363***	(0.040)	−0.311***	(0.018)	ΔP_{t-1}	−0.332***	(0.034)	−0.322***	(0.023)
ΔP_{t-2}	−0.188***	(0.030)	−0.131***	(0.015)	ΔP_{t-2}	−0.172***	(0.028)	−0.132***	(0.016)
ΔP_{t-3}	−0.120***	(0.035)	−0.088***	(0.015)	ΔP_{t-3}	−0.058*	(0.032)	−0.021	(0.020)
$\Delta P_t^{(j-2)}$	0.064**	(0.025)	0.020	(0.010)	$\Delta P_t^{(j-2)}$	0.093***	(0.027)	0.053***	(0.014)
$\Delta P_t^{(j-1)}$	0.063**	(0.025)	0.051***	(0.012)	$\Delta P_t^{(j-1)}$	0.212***	(0.034)	0.168***	(0.021)
$\Delta P_t^{(j+1)}$	0.144***	(0.034)	0.136***	(0.016)	$\Delta P_t^{(j+1)}$	0.109***	(0.025)	0.140***	(0.024)
$\Delta P_t^{(j+2)}$	0.049**	(0.024)	0.036***	(0.011)	$\Delta P_t^{(j+2)}$	0.021	(0.030)	0.042	(0.016)
ρ^{Auc}	0.000	(0.015)	0.002	(0.006)	ρ^{Auc}	−0.011	(0.021)	0.006	(0.007)
V_t	−0.025***	(0.006)	−0.017***	(0.003)	V_t	−0.022*	(0.010)	−0.007	(0.007)
Δw_t^n	−0.528	(0.378)	−1.135***	(0.208)	Δw_t^n	−0.564	(0.280)	−1.179***	(0.202)
Δw_t^p	−0.554	(0.421)	−0.688**	(0.247)	Δw_t^p	−0.390	(0.346)	−0.228	(0.188)
Δs_t^n	0.852	(0.362)	−1.604***	(0.417)	Δs_t^n	0.181	(0.344)	−1.750***	(0.374)
Δs_t^p	−2.842***	(0.513)	−1.860***	(0.403)	Δs_t^p	−2.629***	(0.398)	−1.485***	(0.376)
Δt	0.098***	(0.029)	0.058***	(0.016)	Δt	0.029	(0.029)	0.027	(0.016)
#Obs	3131		9922		#Obs	2759		9284	
R^2_{adj}	0.201		0.190		R^2_{adj}	0.188		0.198	

H13Q3					H13Q4				
variable	regime 1		regime 2		variable	regime 1		regime 2	
	$\xi^{**} \leq 1.324$		$\xi^{**} > 1.324$			$\xi^{***} \leq 0.995$		$\xi^{***} > 0.995$	
	estimate	s.e.	estimate	s.e.		estimate	s.e.	estimate	s.e.
const	−0.687	(0.390)	−1.215	(0.986)	const	−0.396	(0.541)	−0.183	(0.377)
ξ	1.016	(0.497)	0.707	(0.649)	ξ	−0.111	(0.800)	−0.196	(0.204)
ΔP_{t-1}	−0.287***	(0.024)	−0.306***	(0.023)	ΔP_{t-1}	−0.331***	(0.027)	−0.298***	(0.019)
ΔP_{t-2}	−0.143***	(0.024)	−0.165***	(0.021)	ΔP_{t-2}	−0.176***	(0.025)	−0.135***	(0.014)
ΔP_{t-3}	−0.058*	(0.018)	−0.080***	(0.017)	ΔP_{t-3}	−0.095***	(0.023)	−0.071***	(0.013)
$\Delta P_t^{(j-2)}$	0.061***	(0.017)	0.044	(0.017)	$\Delta P_t^{(j-2)}$	0.038	(0.018)	0.043***	(0.013)
$\Delta P_t^{(j-1)}$	0.059***	(0.016)	0.147***	(0.031)	$\Delta P_t^{(j-1)}$	0.114***	(0.021)	0.168***	(0.022)
$\Delta P_t^{(j+1)}$	0.147***	(0.021)	0.230***	(0.029)	$\Delta P_t^{(j+1)}$	0.069***	(0.021)	0.062***	(0.013)
$\Delta P_t^{(j+2)}$	0.075**	(0.019)	0.044	(0.014)	$\Delta P_t^{(j+2)}$	0.045**	(0.018)	0.061***	(0.014)
ρ^{Auc}	−0.016	(0.017)	−0.004	(0.008)	ρ^{Auc}	0.011	(0.014)	0.010*	(0.006)
V_t	0.012*	(0.006)	0.014	(0.004)	V_t	0.012*	(0.005)	0.013***	(0.003)

(Continued.)

Table 2. (Continued.)

H13Q3					H13Q4				
variable	regime 1		regime 2		variable	regime 1		regime 2	
	$\xi^{**} \leq 1.324$		$\xi^{**} > 1.324$			$\xi^{***} \leq 0.995$		$\xi^{***} > 0.995$	
	estimate	s.e.	estimate	s.e.		estimate	s.e.	estimate	s.e.
Δw_t^n	−1.029	(0.333)	−0.753***	(0.205)	Δw_t^n	−1.473***	(0.396)	−1.188***	(0.231)
Δw_t^p	−0.094	(0.184)	−0.594**	(0.261)	Δw_t^p	−0.366	(0.329)	−0.407	(0.222)
Δs_t^n	0.133	(0.355)	−2.276***	(0.437)	Δs_t^n	−0.211	(0.377)	−2.403***	(0.406)
Δs_t^p	−1.967***	(0.319)	−0.943	(0.474)	Δs_t^p	−1.232	(0.481)	−0.259	(0.496)
Δt	−0.011	(0.021)	0.027	(0.021)	Δt	−0.068*	(0.028)	−0.031	(0.019)
#Obs	5626		7190		#Obs	3893		11261	
R^2_{adj}	0.129		0.214		R^2_{adj}	0.144		0.182	

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

positive wind forecast changes Δw_t^p are only significant and negative in the steep regime for Q1 and Q3. In hour H7, the coefficients of Δw_t^p are significant and negative in the flat regime for all contracts as well as in the steep regime for Q1 and Q2, while in hour H18, they are significant and negative in both regimes for Q3. The negative coefficients imply that electricity prices decline in the wake of rising wind power infeed, which is consistent with our intuition. No asymmetry in the coefficients of wind forecast changes between the regimes is observed. We conclude that generally forecast errors of wind power generation contribute to pricing intraday electricity contracts.

The coefficients of negative solar forecast changes Δs_t^n are significant and negative in the steep merit-order regime, but not significant in the flat regime for contracts in hours H13 and H7. In hour H18, the coefficients of Δs_t^n are significant in both regimes for Q3. In the steep regimes, the coefficients of Δs_t^n are at least two times larger by absolute value than in the flat regimes. This reflects the fact that renewable forecast changes affect electricity prices more severely in the steep than in the flat merit-order regime. In hour H13, the coefficients of positive solar forecast changes Δs_t^p are significant and negative in both regimes for contracts Q1 and Q2. In hour H7, the coefficients of Δs_t^p are significant and negative in both regimes for Q4, while in hour H18, they are generally not significant. The negative coefficients of both negative and positive forecast errors of solar power production are reasonable: a lower expectation of solar power infeed should increase electricity prices, whereas a larger prediction of solar power decreases electricity prices.

Generally, renewable forecast changes are more significant in the steep than in the flat merit-order regime. When the market is in the steep merit-order regime, market participants rely on the use of more expensive power generation technologies, which puts additional pressure on them to balance the volatile renewable energies on the intraday market.

Eventually, intraday trading of 15-min contracts is driven by both price-related and fundamental variables at noon, while the price-related variables have a slight surplus importance. For morning and evening contracts, however, intraday price changes are predominantly influenced by past, idiosyncratic price information and the price information of neighbouring contracts.

5. Conclusion

This paper develops an econometric price model with fundamental impacts for intraday electricity markets of 15-min contracts. We analyse a novel and unique dataset of high-frequency transaction data, fundamental supply and demand data, and intradaily updated forecasts of wind

and solar power generation. The nature of our dataset allows the model specification to solely include *ex-ante* market information.

We perform an empirical exploration of transaction prices and trading volumes of 15-min contracts. Empirical evidence suggests that, on average, transaction prices of 15-min contracts exhibit a sawtooth-shaped and trading volumes a U-shaped hourly seasonality. Moreover, liquidity increases sharply within the last trading hour before gate closure. Our empirical analysis also indicates that renewable forecast updates are reflected in intraday prices within one trading minute.

We refine the econometric model by [2] along three dimensions by incorporating: (i) the slope of the merit order curve, (ii) price changes of neighbouring 15-min contracts, (iii) the 15-min intraday auction price. We calibrate our econometric model to market data for a selection of morning, noon, and evening contracts. A threshold regression model is used to examine how 15-min intraday trading depends on the slope of the merit order curve.

Our estimation results reveal that autoregressive price changes up to the third order are highly statistically significant and negative, independent of the time of day. This behaviour provides clear evidence of mean reversion in the price formation mechanism of 15-min contracts. Additionally, price changes of neighbouring contracts exhibit strong explanatory power and a positive impact on price changes of a given 15-min contract. We observe an asymmetric effect of positive and negative renewable forecast changes on intraday prices depending on the merit-order-curve slope: Renewable forecasts affect electricity prices more severely in the steep than in the flat merit-order regime. In general, renewable forecast changes have a higher explanatory power for pricing noon than morning and evening contracts, but price information is the key driver of 15-min intraday trading. Overall, we conclude that the importance of influencing factors on the intraday electricity market has changed from fundamental towards trade-related factors.

As our econometric model exclusively involves *ex-ante* market knowledge, it allows to develop trading strategies for intraday electricity markets, tailor-made for each contract. Furthermore, it helps to design forecasting models for single intraday transaction prices in the continuous trading—to our knowledge, an unexplored territory of scientific research hitherto. Moreover, our article provides a valuable step towards the optimization of the bidding behaviour on intraday markets. Eventually, our insights should prove useful to energy companies within the process of automating intraday electricity trading.

Data accessibility. The intraday transaction data [10,13] and renewable power forecast data [15] are obtained via a (Uni-)Vendor contract and cannot be made publicly available due to contractual conditions. The 15-min intraday auction data may be obtained via [14]. The load data may be downloaded from [16]. Code is publicly available at [21].

Authors' contributions. M.K., R.K. and F.P. conceived of and designed the study; M.K., R.K. acquired the data; M.K. and F.P. prepared the data; M.K. analysed the data, performed the statistical analysis and drafted the manuscript; M.K., R.K. and F.P. critically revised the manuscript. All authors read and approved the manuscript and agree to be held accountable for the work performed therein.

Competing interests. We declare we have no competing interests.

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